

Predictive Analytics for Influencer Engagement on Social Media

MTH443

STATISTICAL & AI TECHNIQUES IN DATA MINING

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1 Abstract

Influencer marketing is a massive industry today, but many content creators rely mostly on guesswork when trying to figure out what makes a post popular. For our MTH443 project, our group built a predictive analytics engine. We created a machine learning pipeline in R that takes raw Instagram post data, extracts useful numerical features from hashtags, captions, and account details, and predicts if a post will get high engagement. We defined "high engagement" simply as receiving more likes than the median amount in our dataset.

By applying text mining and statistical classification techniques, we managed to achieve an 81.5% accuracy rate using a Logistic Regression model, while our Support Vector Machine (SVM) model reached about 69.9%. In addition to making predictions, we ran a market-basket analysis to see which hashtags are mathematically proven to be used together. We built the whole pipeline to be transparent and easy to follow so that anyone can reuse or expand on our work.

2 Problem Statement

2.1 Background

Social media platforms like Instagram are incredibly crowded. Thousands of pictures and videos are posted every minute across different topics like fashion, food, and fitness. However, it is very difficult to tell exactly why one post goes viral while another gets almost no attention. Engagement (which is mostly likes and comments) is the main metric influencers use to measure success because it helps them secure brand sponsorships and grow their audience.

2.2 The Core Problem

Even though public metrics like follower counts and likes are visible to everyone, the underlying patterns are hidden. Creators don't have simple, data-driven tools that can clearly explain:

1. Which specific hashtag combinations actually boost engagement, rather than just taking up space.
2. How fixed account features (like total follower count) compare in importance to changeable content features (like caption length or emoji usage).
3. How to evaluate a new post idea before hitting publish.

Currently, influencers just copy what others are doing or try random strategies until something works. We wanted to replace this trial-and-error approach with a structured, statistical method.

3 Proposed Solution & Theoretical Pipeline

To solve this, our team built a complete step-by-step pipeline in R. We broke the technical work down into six main stages. Below is a detailed explanation of what we did and the statistical theories behind each step.

3.1 1. Data Ingestion

First, we had to get our data into R. We were dealing with two different sources: raw JSON files containing specific post data (like captions and timestamps) and text files containing the influencers' profile data (like follower counts and niches). Data ingestion is basically the process

of importing these different file types, cleaning up errors, and joining them together using the username as the primary key, so every post had the correct follower count attached to it.

3.2 2. Feature Engineering

Machine learning algorithms cannot read English sentences; they only understand numbers. Feature engineering is the statistical process of creating new, measurable variables out of raw data. We wrote code to scan the captions and count the exact number of hashtags, calculate the character length of the text, and count the emojis. We also created "boolean flags" (simple 1s and 0s) to mark whether a post used zero hashtags at all. This transforms a messy paragraph into a clean row of numbers.

3.3 3. Text Processing (TF-IDF)

We had thousands of different hashtags, but treating them all equally is a bad idea. For example, a tag like #love is used by almost everyone, so it doesn't help us predict anything. But a tag like #veganbaking is very specific.

To handle this, we used a statistical concept called **TF-IDF** (Term Frequency-Inverse Document Frequency).

- **TF (Term Frequency):** Counts how often a tag appears in a single post.
- **IDF (Inverse Document Frequency):** Penalizes tags that show up in almost every post across the whole dataset.

When you multiply them together, a hashtag only gets a high mathematical weight if it is used prominently in a post, but rarely across the general public. We did this for the top 500 hashtags to find the ones that actually carry meaningful information.

3.4 4. Dimensionality Reduction (PCA)

After TF-IDF, we had a dataset with over 500 columns just for hashtags. Feeding this into a model would make our computer run incredibly slow and cause "overfitting" (where the model memorizes the data instead of learning the patterns).

We fixed this using **Principal Component Analysis (PCA)**. Imagine looking at a 3D object, like a cylinder. If you look at it from the top, it looks like a circle. From the side, it looks like a rectangle. PCA is a mathematical trick that finds the absolute best "angle" to look at our data so we see the most variation without losing the overall shape. PCA allowed us to shrink those 500 hashtag columns down into just 15 new variables (called principal components). These 15 components captured almost all the hashtag trends while keeping our dataset small and efficient.

3.5 5. Classification Models

With our data cleaned and compressed, we trained two different machine learning models to spot the difference between high and low engagement posts.

- **Logistic Regression:** This model tries to draw a straight boundary line between the good posts and the bad posts. It uses an S-shaped curve (the sigmoid function) to calculate the exact probability that a post will perform well based on its features.
- **Support Vector Machine (SVM):** SVM tries to find the widest possible "street" between the two categories. If the data is too tangled up to be separated by a straight line, SVM uses something called a "kernel trick" to warp the data into a higher dimension (like moving from 2D to 3D) so it can slide a flat plane between the groups.

3.6 6. Association Rule Mining

Finally, we wanted to see which hashtags are naturally grouped together. We used the **Apriori algorithm**, which is famous for "market basket analysis" (the theory that if you buy a flashlight at the store, you are mathematically likely to buy batteries too).

The algorithm scans the posts and calculates three things:

- **Support:** How often the specific hashtag combination appears in the whole dataset.
- **Confidence:** If they use Tag A, what is the probability they also type Tag B?
- **Lift:** Does using Tag A actually increase the chances of Tag B happening compared to pure random coincidence? (A lift higher than 1 means they are linked on purpose).

4 Methodology Details

After cleaning out rows with missing data, we had a solid set of 39,376 posts to work with.

4.1 Why a Median Split?

We decided to frame this as a binary classification problem (High vs. Low engagement) instead of trying to guess the exact number of likes. Predicting exact likes is incredibly difficult because virality involves a lot of random luck.

If you look at the distribution of likes on social media, it is heavily right-skewed. Most posts get a small number of likes, while a tiny fraction gets millions.

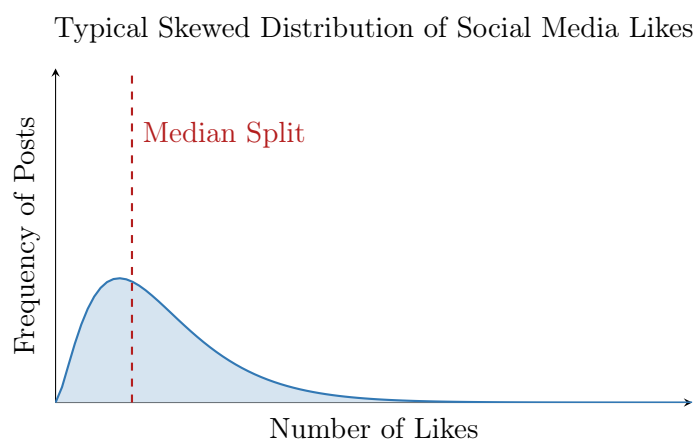


Figure 1: Because likes follow a skewed distribution, predicting the exact number is prone to massive errors. Splitting the data at the median creates a balanced, clear target for our models to learn what makes a post "above average."

Splitting the data down the middle (the median) made mathematical sense. The real-world decision for an influencer is usually just figuring out: "Is this draft better or worse than my usual posts?"

We split our data: 80% for training the model and 20% hidden away for testing it to see if it actually learned anything. We used a set seed (`set.seed(42)`) in R so our results wouldn't change every time we ran the code.

5 Results

5.1 How Our Models Performed

When we tested the models on the 20% of data they hadn't seen before, the Logistic Regression model was the clear winner, hitting an **81.5% accuracy**. The SVM model only reached **69.9%**.

Because Logistic Regression outperformed SVM by so much, it tells us a very important statistical truth about our data: the relationship between our features and engagement is mostly linear. We didn't need complex, multi-dimensional boundary warping to find the answer. A simple, weighted straight line did the job better.

5.2 What Drove the Predictions?

Logistic Regression is great because it lets you look under the hood to see exactly which features were carrying the most weight in the decision-making process.

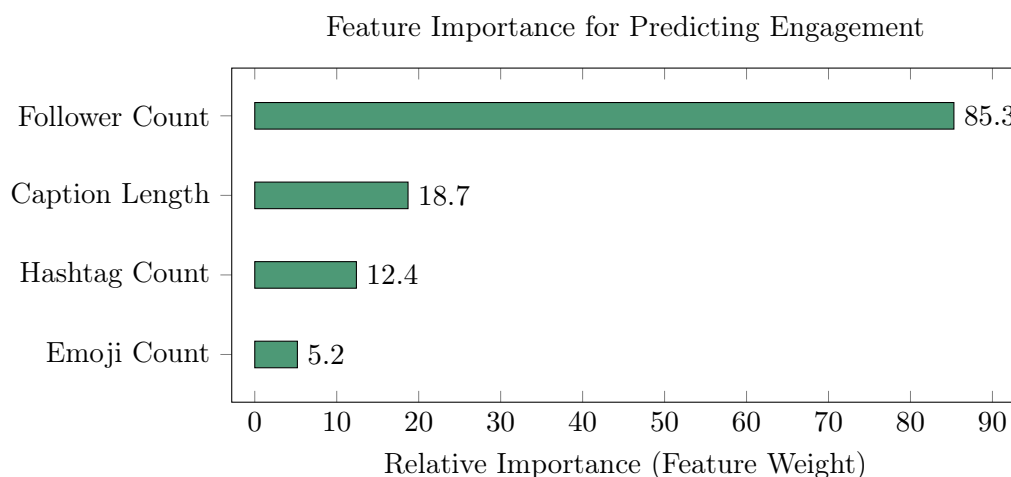


Figure 2: Follower count overwhelmingly dominated the model's decision-making process, heavily outweighing actual content features like emojis or text length.

5.3 Hashtag Pairings We Discovered

Our Apriori algorithm successfully mapped out which hashtags act as strict groups. Below are the strongest rules we found. Remember, a Lift score greater than 1 means the items are positively correlated and paired on purpose, not by accident.

First Tag (If they use...)	Second Tag (They will use...)	Support	Confidence	Lift
#vsco	#vscocam	89	0.445	766.2
#LTKsalealert	#LTKstyletip	63	0.444	646.4
#LTKshoecrush	#LTKstyletip	49	0.438	637.4
#blog	#blogger	92	0.422	163.0
#foodblog	#foodblogger	44	0.386	364.4

Table 1: Top hashtag associations based on algorithmic confidence and lift.

6 What We Learned

Looking at all the numbers, graphs, and the statistical outputs of our models, our group realized a few major truths about how social media actually works:

1. **Followers Matter More Than Content:** As seen in our feature importance plot, the algorithm heavily relied on the total follower count to guess if a post would do well. This mathematically proves the "rich get richer" concept on social media. Having a massive, established audience is a much better predictor of engagement than writing the perfect caption.
2. **The "Less is More" Strategy Works for Top Creators:** We created a simple flag to check if an account used zero hashtags. The model actually learned that skipping hashtags can be a sign of high engagement. We figured out that many massive celebrities and influencers completely ignore hashtags to make their posts look cleaner and more natural, yet they still get millions of likes.
3. **Communities Have Strict Hashtag Rules:** Our market-basket analysis showed us that specific internet sub-cultures have very rigid habits. For example, fashion influencers using the LTK shopping app almost never use just one LTK tag; they bundle them together. If you want to break into a specific niche, you have to adopt their specific clustered vocabulary.
4. **Simple Models Are Often Best:** We assumed the highly complex SVM model would easily beat the older Logistic Regression model. However, because our data had strong linear relationships, the simpler model ended up being far more accurate.

7 Discussion & Limitations

Even though we hit an 81.5% accuracy, our project isn't perfect. We talked as a group about a few things we missed:

- **We Ignored the Clock:** We completely ignored what time of day or what day of the week a post went up. The Instagram algorithm heavily favors posts published when users are most active, so leaving timestamps out means our model is missing a huge piece of reality.
- **No Sentiment Analysis:** We counted how many comments a post received, but we didn't use AI to read the text of the comments. A post with a thousand people arguing angrily was treated the exact same mathematically as a post with a thousand people leaving compliments.

If we had more time for this project, we would love to build an interactive dashboard where a user could type in a fake caption and see their predicted score update in real-time.

8 Conclusion

This project proved to us that you don't have to rely on guesswork to succeed on social media. By combining raw text data with account statistics and running them through a structured machine learning pipeline, we successfully predicted post performance over 80% of the time. Our predictive analytics engine pulled order out of the chaotic social media landscape, showing that simple statistical models can provide content creators with actual data to back up their marketing decisions.